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Tea leaf disease recognition using attention convolutional neural network and handcrafted features

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ABSTRACT

The diseases of tea leaves have a significant impact on their quality and yield, making the rapid identification of leaf diseases in tea crucial for prevention and control. We propose an LBPAttNet model, incorporating a lightweight coordinate attention mechanism into ResNet18 to enhance disease localization and reduce background interference. Furthermore, we employ the local binary patterns (LBP) algorithm to further extract local structural and textural features of tea leaf diseases, and integrate deep features to obtain a more comprehensive feature representation. Additionally, we utilize the focal loss function to alleviate the issues of class imbalance and varying difficulty levels in tea leaf disease, thereby further enhancing the accuracy of tea disease recognition. Our model achieves an accuracy of 92.78% and 98.13% on two publicly available tea disease datasets, surpassing ResNet18 by 3.84% and 2.59% respectively. Compared to traditional algorithms such as AlexNet, GoogleNet, MobileNet, VGG16, and other tea disease recognition algorithms, our model also shows significant improvements. These results highlight the superior performance and robustness of our model.

1. Introduction

Tea is an important economic crop that plays a significant role in agricultural economies. However, during the growth process, tea plants are highly susceptible to various diseases, which can significantly reduce the quality and yield of tea leaves, leading to substantial economic losses (Abade et al., 2021; Mukhopadhyay et al., 2021; Liu et al., 2022a; Zhou et al., 2024). Tea disease detection is crucial for ensuring the economic benefits and product quality of the tea industry, but it faces challenges such as difficulties in early diagnosis, a variety of diseases, and environmental impacts. Traditional methods for identifying tea diseases primarily rely on periodic manual inspections conducted by experts or farmers in tea gardens. However, these methods suffer from subjectivity, inconsistency, and time consumption (Latha et al., 2021; Hu et al., 2021). Hence, there is a pressing need to explore rapid and accurate automated methods for detecting tea diseases.

Traditional methods for detecting plant leaf diseases mainly rely on handcrafted feature extraction techniques, such as HOG (Dalal and Triggs, 2005), SIFT (Lowe, 2004), LBP (Rahim et al., 2013). Suprayoga et al. (2023) utilized HOG technique for feature extraction of mango leaves and employed the XGBoost algorithm for classification. This study successfully classified mango leaf diseases, demonstrating the effectiveness of combining XGBoost algorithm with HOG feature extraction. Aruraj et al. (2019) employed LBP to extract texture features of banana plant diseases and input the extracted features into support vector machine (SVM) for classification, achieving classification accuracy of 89.1% and 90.9% for two types of banana plants respectively. Bashir et al. (2019) employed the SIFT technique to extract features from rice leaf images and utilized SVM-based image processing approach for analysis and classification, achieving an accuracy of 94.16%, thereby providing robust support for farmers in crop protection. Handcrafted feature extraction methods possess certain advantages, focusing primarily on extracting low-level visual features which often demonstrate robustness and generalization. Moreover, they are less reliant on sample size but require handcrafted feature extraction. However, in practical applications, the accuracy of handcrafted features may be influenced by factors such as complex image backgrounds, lighting variations, and viewing angles, which can reduce classification accuracy.

Over the past few years, deep learning methods have been widely applied in the classification of plant leaf diseases and have achieved significant results in numerous pest detection applications (Waheed et al., 2020; Zhao et al., 2024; Wang et al., 2024; Nagachandrika et al., 2024; Xu and Li, 2024). Hassan and Maji (2022) proposed a novel deep learning model based on the Inception layer and residual connection, and was tested on three different plant diseases. The proposed model

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achieved higher accuracy than existing deep learning models. Additionally, Latif et al. (2022) proposed an improved transfer learning method based on VGG19, which accurately detected and diagnosed six different rice diseases, with an improved accuracy rate of 96.08%. Elfatimi et al. (2022) utilized MobileNetV2 to tested on a soybean disease dataset, achieving an accuracy of over 92%. Furthermore, Ozguven and Adem (2019) utilized the Faster R-CNN model to estimate the severity of sugar beet leaf diseases, achieving an overall correct classification rate of 95.48%. Compared to traditional handcrafted feature extraction methods, deep learning methods can capture higher-level semantic information while avoiding the cumbersome process of feature selection (Peng et al., 2021). However, neural networks require a large amount of data for training, and their performance depends largely on training strategies, optimization techniques, preprocessing, and other factors. Additionally, deep learning may result in the loss of some important low-level information, such as edges and contours.

Considering the complementary nature of handcrafted features and deep features, the fusion of these two types of features is a method to improve disease recognition. Recent experimental results have shown that combining deep feature with handcrafted feature typically leads to significant performance improvements (Kumari and Anand, 2023; Ren et al., 2019). Hosny et al. (2023) proposed a lightweight model that combines deep convolutional neural networks with LBP feature fusion for the efficient and accurate classification of various plant leaf diseases. Handcrafted features can describe the underlying visual information of images and possess good interpretability, while deep features depend on training data and have poorer interpretability but are effective in capturing abstract semantic feature information from images. Although the method proposed by Hosny et al. achieved promising results, the LBP algorithm primarily focuses on texture features and fails to adequately emphasize key local features relevant to lesion localization. This limitation may reduce its effectiveness in identifying lesions within complex backgrounds. Furthermore, existing approaches lack effective strategies to address class imbalance, potentially limiting their performance in tea disease recognition tasks.

To address the challenges of diverse tea leaf diseases and limited sample data, using ResNet18 as a relatively shallow backbone for feature extraction can help avoid overfitting that may occur with deeper networks. Additionally, deep learning algorithms may fail to allocate attention weights accurately, thus overlooking key areas of tea leaf diseases against complex backgrounds and reducing recognition accuracy. Introduction of coordinate attention mechanism can help reduce background interference and enhance the model's feature extraction capabilities. Meanwhile, LBP can capture subtle texture changes in tea leaf diseases, making it effective for detecting symptoms such as spots and patches in small affected areas.

Inspired by the aforementioned works, we propose the LBPAttNet model. This model effectively enhances the network by incorporating an attention mechanism to extract high-level semantic features of tea leaf diseases. We also integrate the shallow features of tea leaf diseases extracted by LBP. Next, we fuse these two feature types and use the softmax function at the output layer for classifying tea leaf diseases. We also address sample imbalance using the focal loss function, thereby improving the recognition accuracy of the model.

Compared with existing methods, the key contributions of this paper are:

1. This paper introduces a novel approach to tea disease identification by combining deep features extracted via optimized CNN with handcrafted features.
2. Introducing the attention mechanism into ResNet18 allows for greater focus on the diseased areas of tea leaves, reducing background interference and improving the model's ability to extract features from tea leaf diseases.
3. The LBP algorithm effectively extracts texture features of tea diseases, offering robust interpretability and generalization. Meanwhile, deep learning excels at capturing high-level semantic features. Integrating these two types of features results in a more comprehensive and varied feature representation.
4. On the two publicly available tea leaf disease datasets, our model achieves accuracies of 92.78% and 98.13%. Its overall superior performance compared to other deep learning algorithms demonstrates the effectiveness of the approach.

2. Related works

2.1. Tea leaf disease recognition

Recently, significant progress has been made in the field of tea disease recognition, which can be categorized into two major research directions: utilizing handcrafted feature extraction combined with machine learning algorithms for disease identification, and employing deep learning techniques to automatically extract features and recognize diseases. Hossain et al. (2018) selected 11 key features and successfully used SVM to identify two major tea diseases, achieving an accuracy of over 90%. Sun et al. (2019) proposed an algorithm combining simple linear iterative clustering (SLIC) with SVM, which enhanced the ability to identify tea leaf diseases in complex backgrounds. However, the performance of SLIC largely depends on the selection of parameters for the initial number of cluster centers and has limited adaptability to complex backgrounds. Mukhopadhyay et al. (2021) utilized PCA and multi-class SVM for feature extraction and recognition, achieving an average accuracy of 83% across five major tea diseases. However, the accuracy was lower for diseases with smaller sample sizes. Srivastava and Venkatesan (2020) proposed a texture-based image processing method that extracted texture features using the GLCM and combined them with color features for classification. Bhagat and Kumar (2024) employed deep learning to extract features and build models using various machine learning classifiers. However, deep learning models may learn a large number of high-dimensional, redundant, or irrelevant features. Hu et al. (2019) proposed an improved convolutional neural network (CNN) that integrates multi-scale feature extraction modules and depthwise separable convolution techniques, enhancing the automatic feature extraction capability for tea disease images. Zhang et al. (2023) utilized the IEM-ViT model combined with information entropy weighting and masked autoencoders, achieving a high accuracy of 93.78% in recognizing seven types of tea diseases, demonstrating its potential for diversified disease identification. However, the model's complexity may require more computational resources during training and inference. Soeb et al. (2023) collected images of five types of leaf diseases from tea plantations and applied YOLOv7 to study tea diseases in natural scenes, achieving results that outperformed existing methods in object detection and recognition. Datta and Gupta (2023) proposed a tea disease detection method, constructing a CNN model with multiple convolutional and fully connected layers, which successfully classified healthy and diseased tea leaves. Bhuyan et al. (2024) significantly improved tea disease recognition performance by leveraging enhanced feature extraction and attention mechanism. However, their study was limited to testing two diseases on a single dataset.

Traditional methods based on handcrafted feature extraction perform reliably on small-scale datasets with low computational requirements. However, they rely heavily on manually selected features, lack adaptability to complex backgrounds and diverse diseases. In contrast, deep learning methods can learn complex features, offering higher accuracy and adaptability. Nevertheless, they require substantial data and computational resources and are prone to issues such as overfitting and redundant features. In summary, while significant progress has been made in tea disease recognition, challenges remain, including background complexity, limited data availability, and insufficient model generalization. Therefore, improving recognition accuracy and robustness remains a critical issue to be addressed in this field.

2.2. The application of attention mechanism in plant leaf disease recognition

Although efforts have been made in the field of tea disease recognition, there is still considerable room for improvement. In recent years, methods that combine attention mechanism and deep learning have been effective in reducing background interference and enhancing feature extraction capabilities, achieving notable results. Liu et al. (2022b) proposed a crop disease recognition method based on an improved lightweight CNN, which incorporates a coordinate attention mechanism. Experimental results demonstrated that the model achieved an accuracy of 91.94% on the AI Challenger 2018 plant disease recognition dataset. Bhuyan et al. (2023) introduced an improved deep convolutional neural network, which combines the convolutional block attention module (CBAM), enabling the precise extraction of complex features associated with various diseases. Zhao et al. (2021) developed an enhanced CNN that integrates SeNet, significantly improving its ability to identify leaf diseases. The model achieved high accuracy rates of 96.81% and 99.24% for tomato and grape leaf disease recognition, respectively. These results indicate that the inclusion of attention modules not only accurately captures the complex features of diseases but also does not substantially increase the model's parameter count. Li and Wang (2024) proposed a method based on ResNet combined with the efficient channel attention (ECA) mechanism. Experimental results showed that the ResNet50 model with the ECA mechanism achieved significant improvements in both accuracy and loss. Chen et al. (2021) introduced a lightweight apple leaf disease recognition method based on an improved ShuffleNet V2. By incorporating the BAM attention module, it effectively reduced interference from complex backgrounds, resulting in a 1.5% reduction in error rate and significantly improving the accuracy of apple leaf disease recognition.

Although the attention mechanism has significantly improved the overall performance of plant disease recognition, challenges remain in extracting local texture information and adapting to small sample sizes. Combining traditional handcrafted features with the attention mechanism could provide an effective solution. Integrating both deep learning features and handcrafted features, the model's performance can be further enhanced.

3. Method

This paper proposes a method that combines handcrafted features with deep learning features, as illustrated in Fig. 1. Initially, tea disease images are input, and deep feature Z_C are extracted using a ResNet18 network with attention mechanism. Simultaneously, the LBP algorithm is employed to extract texture features of tea disease, obtaining corresponding histogram feature Z_H. Next, the deep feature Z_C and handcrafted feature Z_H are fused to obtain F_a, which passes through a fully connected (FC) layer to generate feature F_b. Finally, the Softmax function is applied to determine the corresponding categories of tea disease features.

3.1. Feature extraction module based on attention mechanism

ResNet18 has lower computational complexity and resource requirements, while its residual structure effectively alleviates the vanishing gradient problem, ensuring stable training and performance. Compared to other complex models like DenseNet and EfficientNet, ResNet18 achieves efficient feature representation while avoiding high memory overhead and computational costs, making it more suitable for handling complex backgrounds and diverse features in tea disease images. ResNet18 is chosen as the feature extraction network for tea leaf disease recognition to ensure strong performance and prevent gradient vanishing during training. However, since different diseases often exhibit features in specific regions of the image, an attention mechanism is necessary to enhance the model's ability to focus on these critical areas. The attention mechanism can focus the network's attention on target regions, reducing interference from irrelevant information and improving network performance. In this study, the feature extraction capability of the model is enhanced by introducing a coordinate-based attention mechanism. The architecture of the coordinate attention mechanism is illustrated in Fig. 2.

To enable the attention mechanism to capture positional information and establish long-range spatial dependencies, pooling kernels of size $(H, 1)$ and $(1, W)$ are used in both horizontal and vertical directions to encode each channel of the input $X = \{x, x_2, \ldots, x_c\} \in R^{C\times H\times W}$. The output $z_c^h(h)$ obtained for the cth channel at height h, as well as the output $z_c^w(w)$ for the cth channel at width w,

$$z_c^h(h) = \frac{1}{W}\sum_{0\le i<W} x_c(h,i), \tag{1}$$

$$z_c^w(w) = \frac{1}{H}\sum_{0\le j<H} x_c(j,w). \tag{2}$$

Concatenate the feature maps of $z_c^h(h)$ and $z_c^w(w)$, then perform feature transformation using the 1×1 convolutional function $F1$ to obtain the intermediate feature map f in the spatial and vertical directions, where δ is the non-linear activation function,

$$f = \delta(F1([z^h, z^w])), f \in R^{C/r\times(H+W)}. \tag{3}$$

Decompose the feature map f into f_h and f_w along the horizontal and vertical directions, respectively. Then, restore the number of channels of the feature map to be the same as that of the input feature X using 1×1 convolutional operations F_h and F_w. Finally, obtain the horizontal attention weights g^h and vertical attention weights g^w of the feature map using the sigmoid function,

$$g^h = \sigma(F_h(f^h)), f^h \in R^{\frac{C}{r}\times H}, \tag{4}$$

$$g^w = \sigma(F_w(f^w)), f^w \in R^{\frac{C}{r}\times W}. \tag{5}$$

Finally, the output $y_c(i,j)$ of the cth channel $x_c(i,j)$ of input X after coordinate attention can be represented as,

$$y_c(i,j) = x_c(i,j)\times g_c^h(i)\times g_c^w(j). \tag{6}$$

Tea leaf disease images contain abundant target and background information. In this study, ResNet18 was selected as the feature extraction network for disease identification, and a coordinate attention module was adopted to enhance the network's sensitivity to tea disease spots. The incorporation of the coordinate attention module into the ResNet18 network architecture is illustrated in Fig. 3. ResNet18 consists of 4 residual blocks, with coordinate attention mechanism added after each residual block. This mechanism integrates the feature maps adjusted by the coordinate attention module with the input feature maps of the residual block, enhancing the network's ability to represent image features. This optimization contributes to further improving the model's performance in the task of tea disease recognition.

3.2. Handcrafted features extraction module

LBP is a local shape descriptor used to describe local texture patterns in grayscale images. It is widely applied in various fields such as face recognition and pedestrian detection (Liao et al., 2023). LBP is particularly effective for tea leaf disease detection due to its efficient computation, strong texture description, and insensitivity to lighting and rotation. Compared to more complex methods, LBP is simpler to implement, enhancing both real-time processing capability and system robustness. LBP extracts distinctive texture features from a given image by encoding the relationship between each pixel and its neighboring pixels. The mathematical expression of the algorithm is as follows,

$$LBP(x_c, y_c) = \sum_{i=0}^{P-1} s(g_i - g_c)\times 2^i, \tag{7}$$

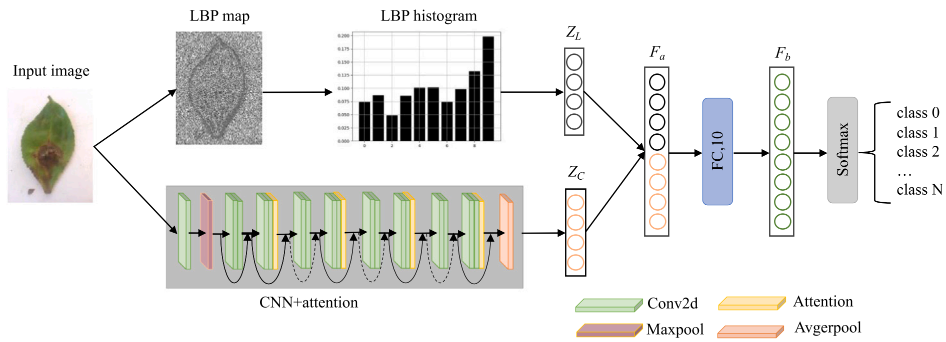

Fig. 1. The architecture of tea disease recognition system. The LBPAttNet utilizes the LBP algorithm to extract texture features of tea leaves disease, followed by computing corresponding histogram features. Deep learning features are extracted using a CNN with an attention mechanism. Here, Z_C denotes the extracted deep learning features, Z_L represents the extracted handcrafted features, F_a signifies the fusion of handcrafted features and deep features, and F_b corresponds to the features obtained after passing through the fully connected (FC) layer.

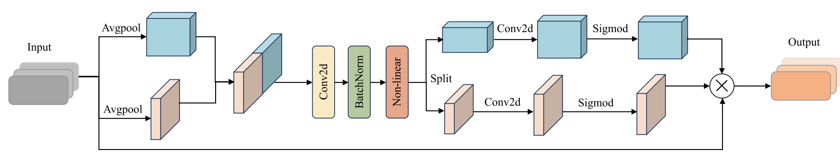

Fig. 2. The architecture of the coordinate attention mechanism. Coordinate Attention integrates position embeddings and coordinate embeddings, considers positional information when calculating attention scores, thus capturing relationships between different positions in the input sequence, and finally obtains the output through weighted summation.

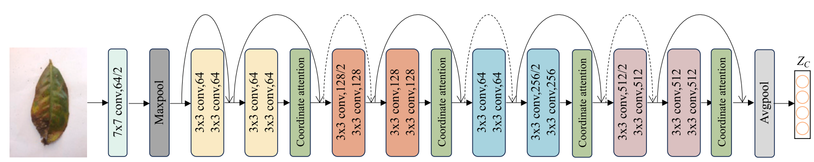

Fig. 3. The architecture of the deep learning feature extraction module. ResNet18 is adopted as the backbone network, and the coordinate attention is incorporated to enhance the model's feature extraction capability.

(x_c, y_c) represents the position of the central pixel in LBP, g_i represents the grayscale value of the central pixel, g_c represents the grayscale values of the neighboring pixels. P is the number of pixels in the neighborhood, and $s(x)$ is a binary function, the formula is,

$$s(x) = \begin{cases} 1, x \geq 0, \\ 0, x < 0. \end{cases} \tag{8}$$

LBP operator generates binary codes which increase with the number of pixels in the neighborhood, leading to increased computational complexity and affecting feature description. To address this issue, this paper introduces a uniform LBP (ULBP) model. It reduces the size of feature vectors, achieves simple rotational invariance, and preserves image features.

$$ULBP(x_c, y_c) = \begin{cases} \sum_{i=0}^{P-1} s(g_i - g_c) \times 2^i, U(LBP(x_c, y_c)) \leq 2, \\ P + 1, other, \end{cases} \tag{9}$$

$U(LBP(x_c, y_c))$ represents the number of transitions from 1 to 0 or from 0 to 1 in LBP. In images of tea disease, the majority of transitions from 1 to 0 and from 0 to 1 are less than or equal to 2. Therefore, adopting ULBP can effectively simplify the feature vector. This paper employs the ULBP algorithm with the optimal radius $R = 1$ and sampling points $P = 1$ to extract feature vector maps of tea disease images. By sliding windows, the ULBP values within each window are computed to construct a histogram of the window. Then, the window is moved to the next region, and the process is repeated. Once histograms for all sliding windows across the entire image are constructed, they are concatenated to form the LBP feature vector map of the image. LBP maps and their corresponding histograms collected for tea diseases are shown in Fig. 4.

3.3. Classification and loss function

The number of samples for tea diseases may have class imbalances, where the classification difficulty between healthy and diseased tea leaf samples varies. The cross-entropy loss treats all samples equally, resulting in suboptimal model training. This paper adopts the focal loss (Lin et al., 2017) function to address this issue, as shown in the following formula,

$$FL(p_t) = -\alpha_t (1 - p_t)^{\gamma} \log(p_t), \tag{10}$$

p_t represents the predicted probability of tea disease, while parameters α and γ are two hyperparameters used to adjust sample weights. Parameter α addresses the issue of imbalanced positive and negative samples, while γ handles the difficulty of samples. When training the model, more attention can be given to the challenging tea disease samples, thereby improving the model's recognition accuracy.

4. Experiments and results

4.1. Datasets introduction and implementation details

The tea leaf diseases dataset used is derived from two publicly available datasets: tea_sickness_disease dataset (Kimutai and Förster, 2022) and tea_leaf_disease dataset (Datta and Gupta, 2023). Tea_sickness_disease dataset contains seven common types of tea leaf diseases: (1) red leaf spot, (2) algal leaf spot, (3) bird eye spot, (4) gray blight, (5) white spot, (6) anthracnose, (7) brown blight. Additionally, it includes images of healthy tea leaf samples. In total, there are eight types of tea leaf samples. Tea_leaf_disease dataset were all taken in

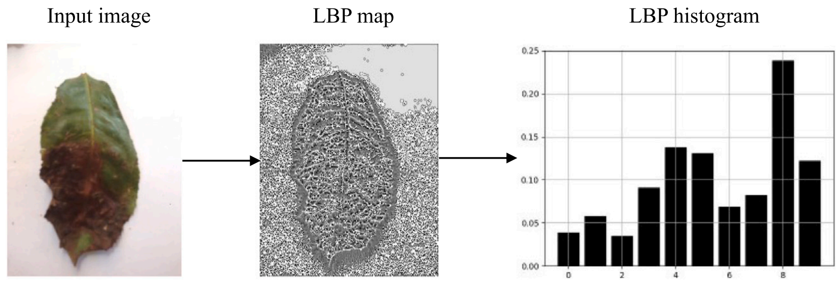

Fig. 4. Handcrafted extraction features of tea disease process. Using the LBP algorithm to extract the texture feature map of tea diseases, followed by the statistical analysis of the corresponding histogram features.

Table 1
The number of samples in the train set and test set for tea_sickness_disease dataset.

Target	Training set	Testing set
Algal leaf spot	56	57
Anthracnose	50	50
Bird eye spot	50	50
Brown blight	56	57
Gray bight	50	50
Red leaf spot	71	72
White spot	71	71
Healthy	37	37

Table 2
The number of samples in the training set and testing set for tea_leaf_disease dataset.

Target	Training set	Testing set
Algal leaf spot	500	500
Brown blight	433	434
Gray bight	500	500
Red leaf spot	500	500
Helopeltis	500	500
Healthy	500	500

natural environments using mobile phone cameras, and disease types were identified with the input of tea plantation experts. The dataset of tea_leaf_disease consists of 5867 images, covering six categories: (1) algal leaf spot, (2) brown blight, (3) gray blight, (4) red leaf spot, (5) helopeltis (6) healthy. To verify the advantage of our algorithm in scenarios with relatively few training samples, the public tea dataset is split into training and testing sets in a 5:5 ratio using random selection, as shown in Tables 1 and 2. To enhance the model's performance and training speed, we uniformly resized the images to 224 × 224 on both public datasets and applied data augmentation techniques such as rotation, translation, and flipping. Notably, data augmentation did not increase the number of images but instead dynamically transformed the images during the data loading phase. The images of tea leaves in the two public dataset are shown in Figs. 5 and 6.

The experimental setup in this study is as follows: Python version 3.11.4, CUDA version 12.2, PyTorch deep learning framework version 2.0.1. The CPU is Intel(R) Xeon(R) Gold 6348 with a clock frequency of 2.60 GHz, and the GPU is NVIDIA GeForce RTX 4090. The proposed algorithm framework utilizes a pre-trained ResNet18 with training images normalized to a size of 224 × 224. The optimizer chosen is Adam, with a training batch size of 16 and a total of 50 epochs. The initial learning rate is set to 1e-5, halving every 20 epochs until reaching a final learning rate of 1e-7.

4.2. Model evaluation

In the task of tea leaf disease identification, accurately assessing the model's performance is crucial. Therefore, it is necessary to select appropriate evaluation metrics to measure the model's performance. This study employs four commonly used evaluation metrics: accuracy, precision, recall, and F1 score, as the measurement standards for evaluating the model. Accuracy is the proportion of correctly identified disease samples to the total number of samples, which is an intuitive evaluation criterion,

$$Accuracy = \frac{TP + TN}{TP + FP + FN + TN}. \tag{11}$$

Precision is the proportion of correctly identified samples for a single disease to the total number of correctly identified samples,

$$Precision = \frac{TP}{TP + FP} \ . \tag{12}$$

Recall is the proportion of correctly identified samples for a single disease to the total number of samples for that disease,

$$Recall = \frac{TP}{TP + FN}. \tag{13}$$

The F1 score is the harmonic mean of precision and recall, reflecting the robustness of the model,

$$F1 = \frac{2 \times TP}{2 \times TP + FP + FN}. \tag{14}$$

True positive (TP) indicates the number of instances where the model predicts tea leaf diseases as positive samples, and they are actually positive samples. True negative (TN) represents the count of instances where the model predicts tea leaf diseases as negative samples, and they are actually negative samples. False positive (FP) signifies the number of instances where the model predicts negative samples as positive for tea leaf diseases. False negative (FN) denotes the count of instances where the model predicts positive samples as negative for tea leaf diseases.

4.3. Results and discussion

The algorithm proposed in this study leverages ResNet18, enhanced by an attention mechanism, as its backbone network. It utilizes LBP for feature extraction and incorporates focal loss to improve network training effectiveness. To further validate the algorithm's performance, we compare the results of the ResNet18 with our proposed LBPAttNet on the two publicly tea leaf disease datasets. We use F1 score and precision as the evaluation metrics to compare the model results.

Fig. 7 illustrates the test results of our model and ResNet18 in terms of the F1 score for tea_sickness_disease dataset. We observe that our algorithm consistently achieves an F1 score equal to or higher than the origin ResNet18. Particularly, our algorithm demonstrates significant improvements of 10% in the cases of bird eye spot and 11% gray blight diseases. Compared to ResNet18, our algorithm achieved a 8% improvement in anthracnose and a 3% increase in brown blight. Fig. 8 depicts the precision test results of the algorithm proposed in this study and origin ResNet18 for tea_sickness_disease dataset. We can observe that our algorithm shows a 4% increase in precision for anthracnose and algal leaf spot diseases compared to ResNet18, while brown blight shows a 2% increase in precision. Particularly, there is an improvement of 16% in identifying gray blight disease. Both models exhibit similarly high recognition rates for healthy and red leaf spot diseases.

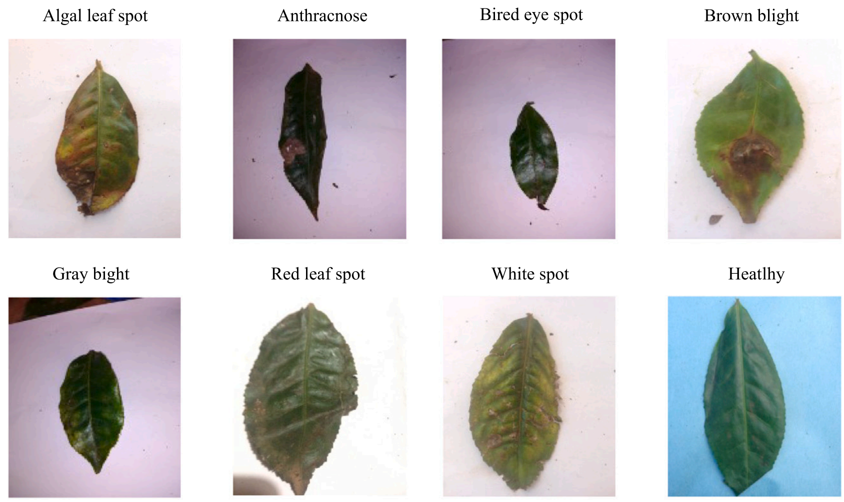

Fig. 5. Tea_sickness_disease dataset images.

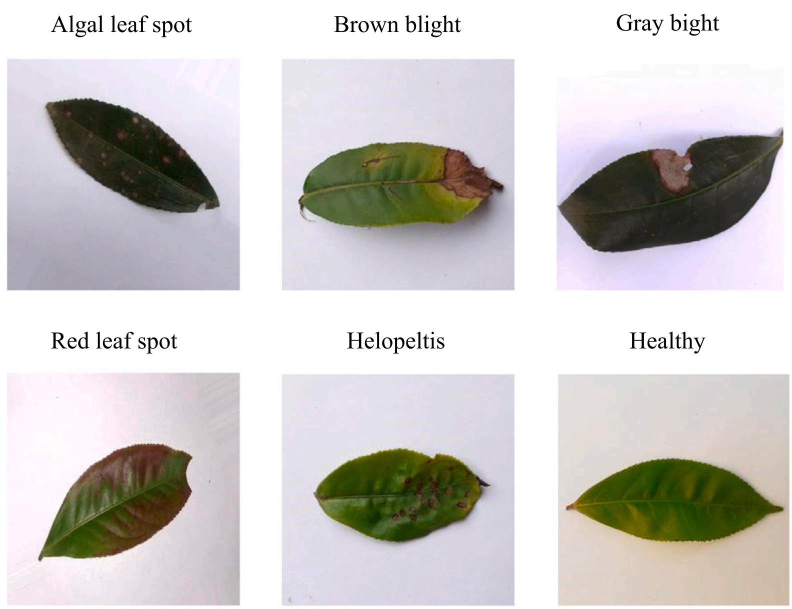

Fig. 6. Tea_leaf_disease dataset images.

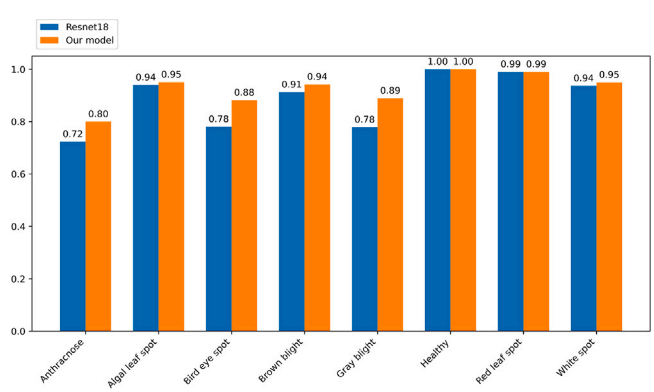

Fig. 7. Comparison F1 score of ResNet18 and LBPAttNet (our model) on tea_sickness_disease dataset.

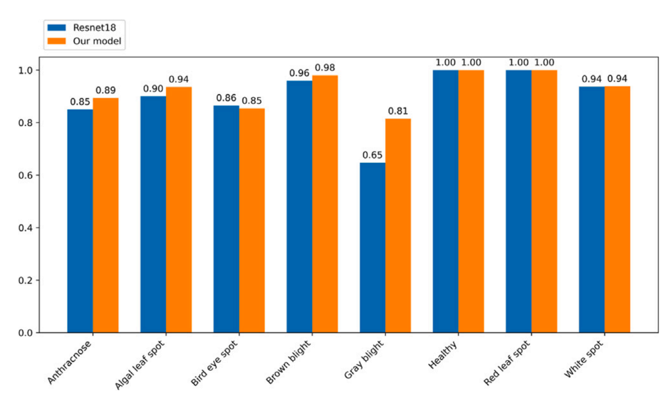

Fig. 8. Comparison precision of ResNet18 and LBPAttNet (our model) on tea_sickness_disease dataset.

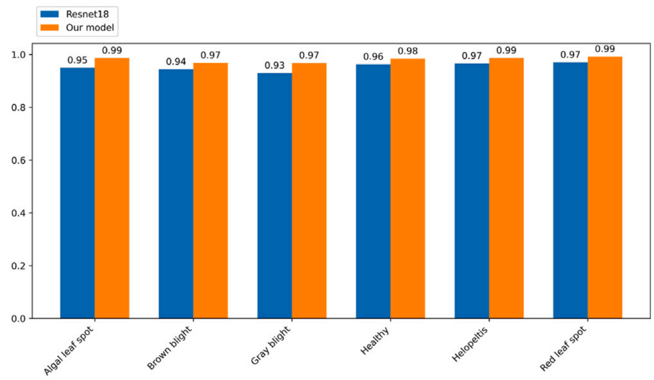

Fig. 9. Comparison F1 score of ResNet18 and LBPAttNet (our model) on tea_leaf_disease dataset.

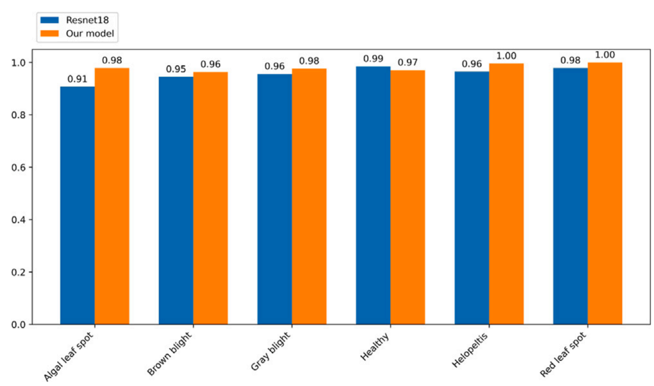

Fig. 10. Comparison precision of ResNet18 and LBPAttNet (our model) on tea_leaf_disease dataset.

We also conducted tests on the tea_leaf_disease dataset. The F1 score results are shown in Fig. 9, and the precision results are in Fig. 10. In terms of F1 score, our algorithm achieves a 4% improvement over the ResNet18 for algal leaf spot and gray blight, and a 3% improvement for brown blight, and also outperforms ResNet18 for other tea leaf diseases. Regarding the precision results, our algorithm achieved a maximum improvement of 7% over ResNet18 for algal leaf spot, and outperformed ResNet18 in identifying all tea leaf diseases except for healthy leaves. Overall, our algorithm demonstrated strong performance on the tea_leaf_disease dataset.

The confusion matrices for our algorithm's performance on two public tea leaf datasets are shown in Figs. 11 and 12. On the tea_sickness_disease dataset, the performance of our model for various diseases is as follows: it correctly identifies red leaf spot 68 times, healthy leaves 33 times, gray blight 44 times, with a maximum of just one misclassification. However, the recognition rate for anthracnose is relatively lower, as these leaf disease types share a high similarity with bird eye spot and gray blight. In summary, while our model performs well in identifying tea leaf diseases, there is room for improvement, especially in classifying certain diseases. On the tea_leaf_disease dataset, for algal leaf spot, our model achieves 508 correct predictions, with 3 instances misclassified as other diseases. Brown blight is correctly identified 426 times, with 11 instances misclassified as gray blight. The model correctly identifies 462 instances of gray blight but misclassifies it as brown blight 15 times. Healthy leaves are correctly identified 495 times, with minimal misclassifications. For helopeltis and red leaf spot, the model correctly classifies 474 and 514 instances, respectively, with few misclassifications. Overall, the model demonstrates high accuracy in classifying most tea leaf diseases, with some confusion primarily between brown blight and gray blight.

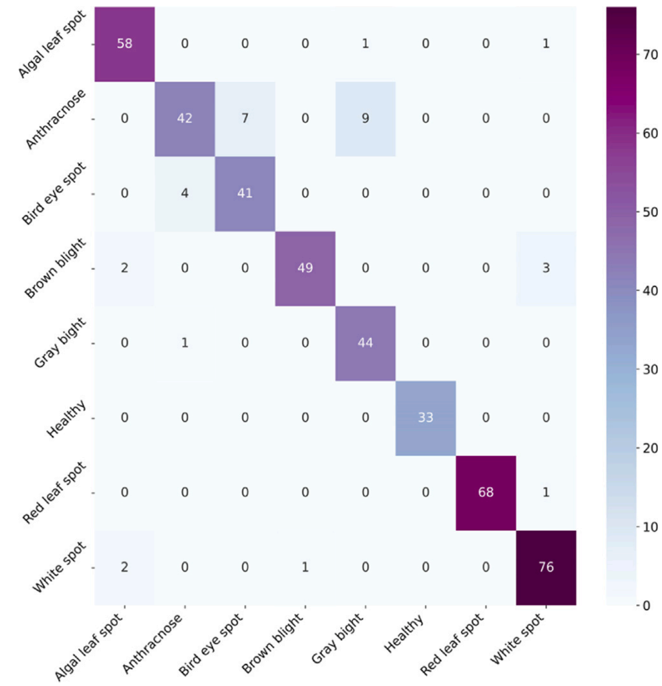

Fig. 11. The confusion matrix of LBPAttNet's predictions on the tea_sickness_disease dataset.

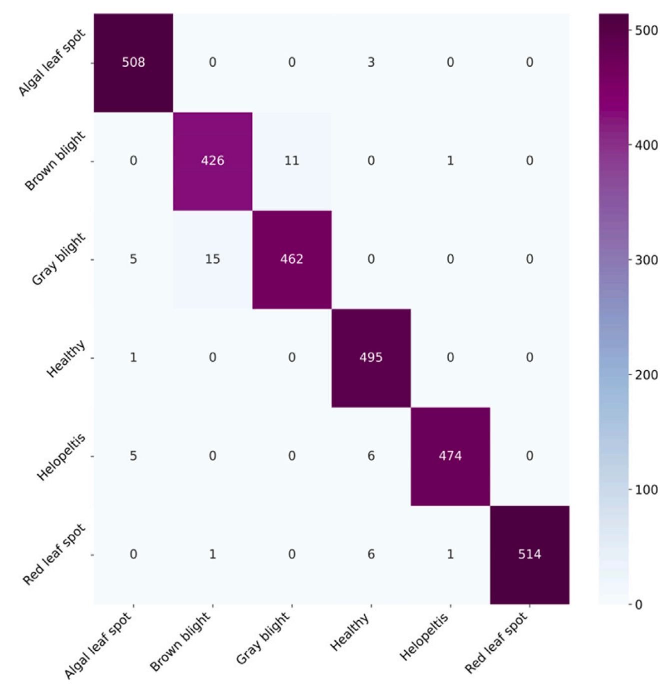

Fig. 12. The confusion matrix of LBPAttNet's predictions on the tea_leaf_disease dataset.

Through this experiment, it clearly shows that our proposed LBPAttNet outperforms the ResNet18 model in terms of overall performance metrics, including F1 score and precision. This observation highlights the clear advantage of our proposed algorithm. By outperforming the baseline ResNet18 model across various evaluation metrics, our LBPAttNet demonstrates its robustness and effectiveness in addressing the challenges of tea leaf disease recognition. Additionally, the superior performance of our algorithm highlights its potential for practical applications in agriculture and emphasizes the significance of incorporating

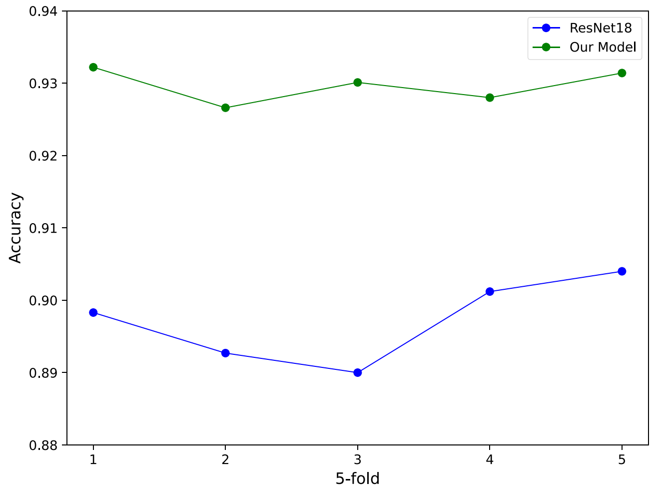

Fig. 13. 5-fold cross validation on tea_sickness_disease dataset.

novel techniques, such as attention mechanisms, handcrafted features and focal loss, into traditional deep learning architectures. Overall, these findings validate the efficacy of our approach and pave the way for future advancements in disease recognition systems for agricultural applications.

To comprehensively evaluate the performance and robustness of the LBPAttNet model, we conducted 5-fold cross validation on the tea_sickness_disease dataset. Fig. 13 illustrates the performance comparison between ResNet18 and our proposed model across the 5-fold cross validation. The experimental results show that our model consistently outperforms ResNet18 in all folds, with an accuracy stabilizing around 0.93 and minimal variation, demonstrating strong robustness and stability. These results further confirm that our improved model not only significantly enhances recognition accuracy but also demonstrates superior stability and generalization capability.

5. Effectiveness of attention mechanism and handcrafted features

To visually demonstrate the capability of attention mechanism in extracting disease features, we selected typical images of tea leaf diseases and used both the ResNet18 and our proposed improved ResNet18 model. In tea leaf images, the model applies weighted processing to different regions through attention mechanisms. Brighter regions in tea leaf images tend to attract more focus from the model, as it seeks features more prominently in these areas to better identify and classify images. Conversely, darker regions receive less attention, as they may not contain significant disease-related information. We employed Grad-CAM (Selvaraju et al., 2017) to visualize the features of the last layer of ResNet18. The visualization results are shown in Fig. 14, it can be observed that in the model without the attention mechanism, the brighter areas are located outside the disease regions, indicating that the model assigns more weight to these regions. However, after introducing the attention mechanism, the brightness of the disease areas on the leaves increases, indicating that the model pays more attention to these disease regions. By incorporating the coordinate attention mechanism, the model effectively captures the relationship between channels and positions. This enables the model to focus more on the positions of tea leaf diseases, reducing attention on irrelevant areas. In this way, the model is able to extract disease-related features more accurately, thereby enhancing its feature extraction capability.

To confirm the enhanced performance due to handcrafted features, we conducted tests on tea_sickness_disease dataset. We visualize the features extracted by our model, which combines handcrafted and deep learning features, as well as those extracted by the ResNet18 model. We use t-SNE (Van der Maaten and Hinton, 2008) for visualization and the results are shown in Fig. 15. The dataset contains 8 classes, including 7 different types of tea leaf diseases and healthy tea leaves. From Fig. 15, it can be observed that the introduction of handcrafted features leads to a significant reduction in intra-class distances among tea leaf disease samples, indicating higher compactness within each disease category. Conversely, inter-class distances become more dispersed, reflecting greater separability between different disease classes. Overall, the observed changes in distance distributions highlight the effectiveness of incorporating handcrafted features in optimizing the feature space and facilitating better disease recognition in tea leaf images. In conclusion, incorporating handcrafted features significantly enhances the distinguishability of leaf diseases by enabling the model to capture more detailed texture information. This improves feature representation and aids in better differentiation of tea leaf disease types.

6. Ablation experiments

To evaluate the effectiveness of the proposed improvement algorithm in enhancing the model's disease recognition performance, this study designed ablation experiments to compare the model's performance with and without the attention mechanism, handcrafted features, and focal loss function. The results on the tea_sickness_disease dataset are presented in Table 3.

Adding the attention mechanism to ResNet18 (mode 1), model 2 achieves a 1.35% improvement in accuracy. Additionally, the F1 score, precision, and recall improve compared to the ResNet18 model, indicating that the attention mechanism enhances the model's feature extraction. Combining handcrafted features with those extracted by ResNet18, model 3 achieves a 1.58% improvement in accuracy compared to the ResNet18 model. The overall performance is also better than model 1, indicating that combining both types of features enhances the model's recognition. Model 4 achieves a 3.16% accuracy improvement compared to model 1 through the integration of the attention mechanism, handcrafted features, and deep features. By combining handcrafted features and focal loss, model 5 achieves significant improvements in various performance metrics compared to model 1. Our model shows a 3.84% increase in accuracy, a 3.23% increase in precision, a 4.42% increase in recall, and a 4.36% increase in F1 score compared to model 1. Focal loss, compared to cross-entropy, adjusts the model's optimization by focusing on hard to classify samples.

Overall, our model shows substantial improvements in performance compared to ResNet18. This highlights the key roles of attention mechanisms, handcrafted features, and focal loss in enhancing the model's capabilities. Leveraging attention mechanisms allows our model to focus on relevant features, improving its discriminative ability. Handcrafted features complement automatic feature extraction, providing extra information to enhance accuracy. Moreover, focal loss enables the model to prioritize hard to classify samples, mitigating the impact of class imbalance and refining performance. These improvements collectively enhance the model's ability to accurately identify tea leaf diseases. By integrating attention mechanism, handcrafted features, and focal loss into the ResNet18 model, we have significantly improved its performance. This highlights the importance of these components in advancing disease recognition systems for agricultural applications.

7. Comparing with other algorithms

In order to comprehensively validate the effectiveness of our proposed model, we conduct comparative experiments with traditional convolutional neural networks. In these experiments, we select four traditional models: AlexNet (Krizhevsky et al., 2017), GoogLeNet (Szegedy et al., 2015), MobileNet (Howard et al., 2017) and VGG16 (Simonyan and Zisserman, 2014), which have achieved remarkable success in the field of image recognition and have been widely applied across

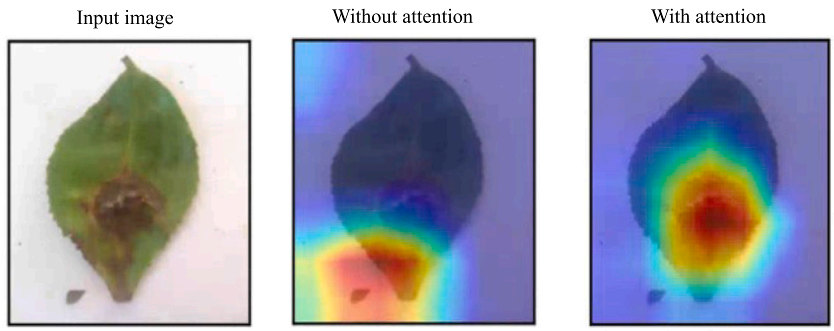

Fig. 14. The feature extraction capability of attention mechanism in models. The image on the left is the original input of a tea leaf disease image. The image in the center is the result obtained from the ResNet18 network, which did not incorporate an attention mechanism. The image on the right is produced by our proposed model, which integrates an attention mechanism. In tea leaf images, brighter regions indicate areas where the model pays more attention, and thus assigns higher weights. Conversely, darker regions receive lower weights as the model allocates less attention to them.

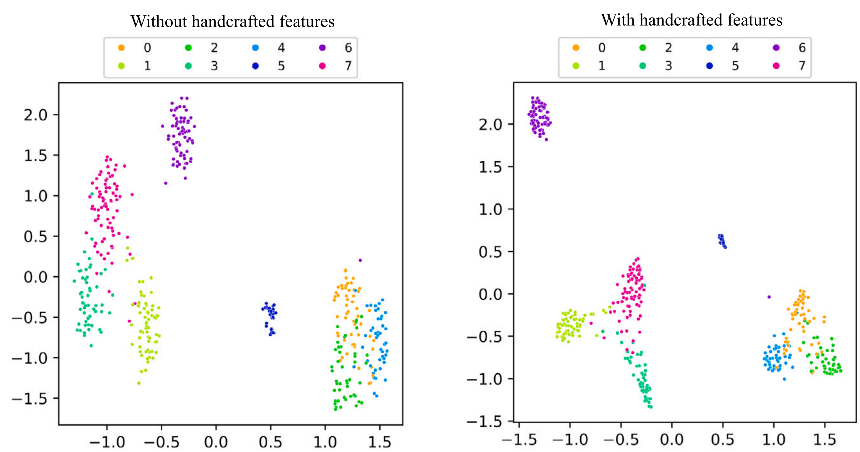

Fig. 15. Comparing the effect of with/without handcrafted features on the tea_sickness_disease dataset. In the visualization feature map, the left side represents the results of origin ResNet18 without handcrafted features, while the right side displays the results of our model with handcrafted features. A smaller intra-class distance indicates that these samples share more similar features. Conversely, a larger inter-class distance reflects that samples from different categories are more distinguishable in feature representation.

Table 3
Ablation experiments of our model on tea_sickness_disease dataset. ×indicates that the method was not used, while ✓indicates that the method was used.

Mode	Attention	Handcarfted features	Focal loss	Accuracy	Precision	Recall	F1 score
Mode 1	×	×	×	0.8894	0.8947	0.8851	0.8822
Mode 2	✓	×	×	0.9029	0.9044	0.9112	0.9021
Mode 3	×	✓	×	0.9052	0.9086	0.9006	0.8990
Mode 4	✓	✓	×	0.9210	0.9185	0.9242	0.9176
Mode 5	×	✓	✓	0.9145	0.9142	0.9133	0.9128
LBPAttNet(Ours)	✓	✓	✓	**0.9278**	**0.9270**	**0.9293**	**0.9258**

various practical scenarios. Additionally, we compare other algorithms for tea leaf disease recognition. On the two publicly tea leaf disease images, we conduct a comparative test between these models and our proposed improved model. The experimental results are shown in Table 4. Under the same experimental conditions, we observe significant improvements in accuracy with our proposed model. On the tea_sickness_disease dataset, compared to AlexNet, GoogleNet, MobileNet and VGG16, our model achieved accuracy improvements of 4.52%, 6.78%, 4.20%, and 4.74%, respectively. Our model also outperforms Datta et al.'s model and Zhang et al.'s model. On the tea_leaf_disease dataset, our model outperforms traditional approaches overall, achieving an maximum accuracy improvement of 4.47%. Compared to the method by Datta et al. our model achieves a 2.46% increase in accuracy. Similarly, when compared to Zhang et al.'s model, it demonstrates an improvement of 1.57%. Runtime is tested on the test sets of two public datasets. Compared to other models, our model achieves superior performance while maintaining a reasonable runtime, demonstrating an excellent balance between performance and efficiency. In addition to accuracy, our model also achieves the best results in precision, recall, and F1 score across both public datasets. This indicates that our model demonstrates superior performance in identifying tea leaf diseases, further validating its effectiveness in tea leaf disease recognition.

Overall, through comparative experiments with traditional models and other tea disease identification algorithms, our improved model demonstrates significant performance enhancements in tea leaf disease recognition tasks, highlighting its greater feasibility and practical value. These findings offer important references and guidance for further advancements in disease recognition research in the agricultural domain.

Table 4
Performance comparison on tea_sickness_disease and tea_leaf_disease dataset.

Dataset	Model	Accuracy	Precision	Recall	F1 score	Running time (s)
tea_sickness_disease	ResNet18 (He et al., 2016)	0.8894	0.8947	0.8851	0.8822	**2.30**
	AlexNet (Krizhevsky et al., 2017)	0.8826	0.8851	0.8841	0.8815	2.34
	GoogleNet (Szegedy et al., 2015)	0.8600	0.8737	0.8623	0.8538	2.45
	MobileNet (Howard et al., 2017)	0.8858	0.9089	0.8850	0.8755	2.38
	VGG16 (Simonyan and Zisserman, 2014)	0.8804	0.8862	0.8850	0.8803	2.58
	Datta and Gupta (2023)	0.8950	0.8958	0.8944	0.8942	2.33
	Zhang et al. (2023)	0.9012	0.9023	0.9001	0.8998	2.77
	LBPAttNet(Ours)	**0.9278**	**0.9270**	**0.9293**	**0.9258**	2.44
tea_leaf_disease	ResNet18 (He et al., 2016)	0.9554	0.9563	0.9545	0.9549	1.74
	AlexNet (Krizhevsky et al., 2017)	0.9499	0.9505	0.9481	0.9486	1.50
	GoogleNet (Szegedy et al., 2015)	0.9366	0.9394	0.9341	0.9349	2.51
	MobileNet (Howard et al., 2017)	0.9482	0.9498	0.9486	0.9477	2.18
	VGG16 (Simonyan and Zisserman, 2014)	0.9523	0.9579	0.9501	0.9515	3.41
	Datta and Gupta (2023)	0.9567	0.9565	0.9559	0.9558	**1.54**
	Zhang et al. (2023)	0.9656	0.9658	0.9647	0.9648	4.65
	LBPAttNet(Ours)	**0.9813**	**0.9810**	**0.9809**	**0.9808**	2.48

8. Conclusions

To address the challenge of prompt identification of tea leaf diseases, this paper proposes a network model that combines handcrafted features with deep convolutional networks enhanced by attention mechanisms for the identification of common tea leaf diseases. Firstly, an attention mechanism module is introduced to reduce background interference, enabling the network to accurately localize target regions and enhancing the feature extraction capability of ResNet18. Secondly, LBP is utilized to extract handcrafted features, enhancing the model's ability to capture finer details. Finally, the two types of features are fused, resulting in a more comprehensive representation of tea leaf disease characteristics. The focal loss function is employed as the loss function to improve recognition of difficult to classify samples, further enhancing the model's recognition capability. Experimental results on two publicly tea leaf disease datasets demonstrate that compared to the ResNet18, the accuracy is improved by 3.84% and 2.59% respectively, indicating superior recognition performance of the LBPAttNet model. When compared with traditional convolutional neural networks and other tea disease identification algorithms, our LBPAttNet model exhibits better recognition performance. While our experiments primarily focus on tea disease identification, the proposed method is equally applicable to other crops and various types of plant diseases. With appropriate adjustments to data preprocessing steps and model parameters based on the specific characteristics of the target crops and diseases, this approach can be effectively applied to a wide range of agricultural scenarios. In certain cases, LBP features may fall short in capturing fine image details, especially when image quality is low or noise levels are high, resulting in weaker robustness of handcrafted features. To address this, we plan to adopt multimodal learning to enhance the model's ability to recognize complex or rare disease samples. For large-scale datasets computational overhead could become a bottleneck. Optimizing the inference process through techniques such as pruning and quantization can effectively reduce computational costs and improve the model's deployment efficiency on resource-constrained devices.

CRediT authorship contribution statement

Peng Wu: Writing – review & editing, Writing – original draft, Validation, Supervision, Software, Resources, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation. **Jinlan Liu:** Methodology, Formal analysis, Investigation. **Mingfu Jiang:** Data curation, Formal analysis, Investigation. **Li Zhang:** Investigation, Validation. **Shining Ding:** Investigation, Data curation. **Kewang Zhang:** Writing – review & editing, Supervision, Conceptualization, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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